

Simulating multi-soliton solutions of the nonlinear Schrödinger equation in Bose-Einstein condensates

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Aims

- Model multi soliton interactions In the 1D Gross-Pitaevskii equation by plotting density distribution found using the semi-implicit finite difference method.
- Investigate soliton behavior over time using the density, analyzing their motions and interactions

Introduction

Bose-Einstein condensation (BEC) [1] occurs for a **dilute boson** gas at extremely low temperatures, causing the bosons to **occupy the lowest energy level**. Gaseous condensates have short **range s-wave** interactions which introduces a nonlinear term into the Schrödinger equation resulting in the **Gross-Pitaevskii equation (GPE)** supporting propagating soliton solutions (Fig 1.). Using the **finite-difference** numerical method we can simulate the solutions for **multi soliton** interactions in a BEC by solving the wavefunction and calculating the density distribution.

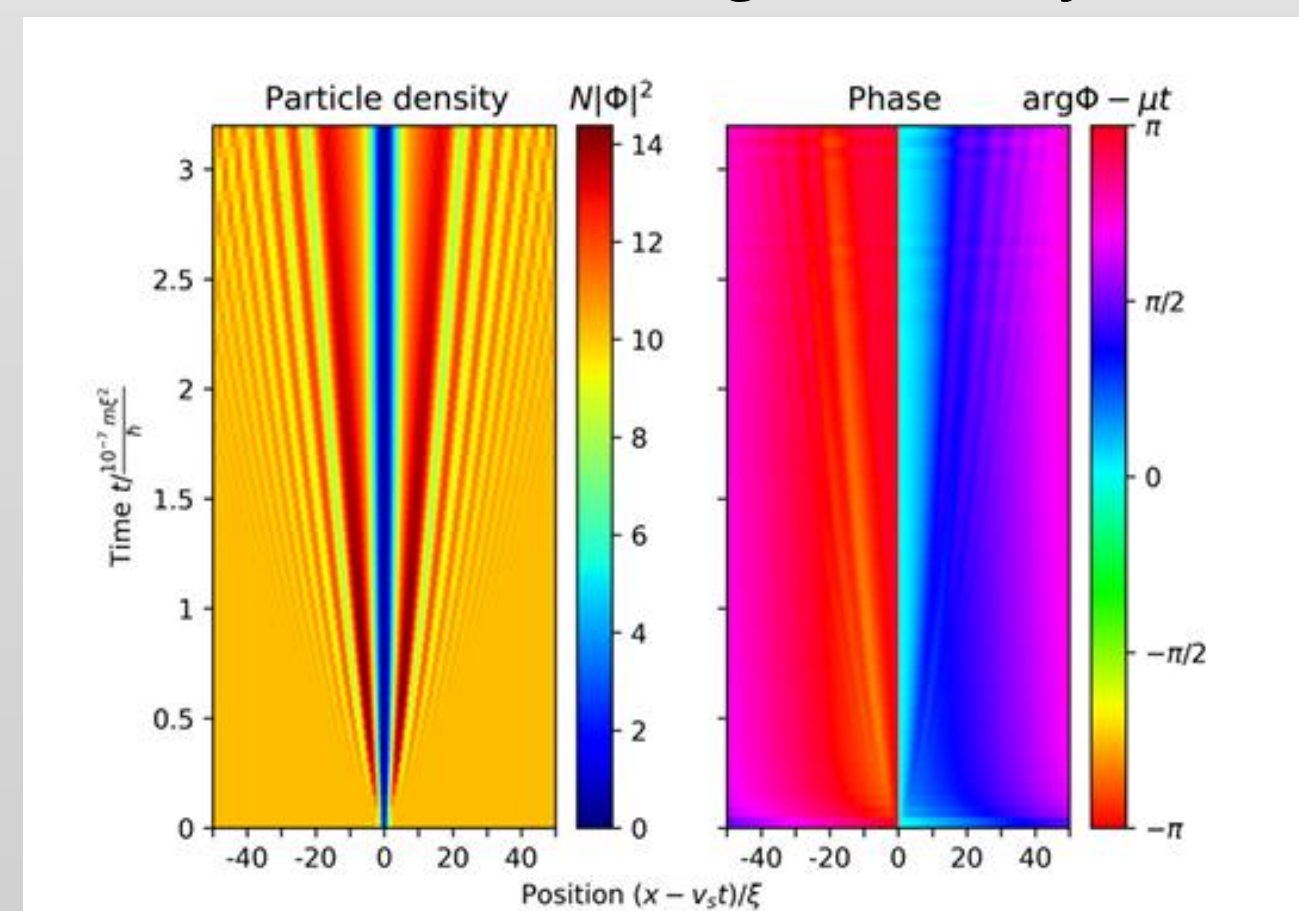


Fig 1. Time evolution of particle density (left) and phase (right) for a soliton of the 1D GPE equation [2]

Soliton Theory

A **soliton** [3] is a wave in which collisions with other solitons **preserve its shape and velocity** as propagation continues. This is caused by their dispersion **being balanced by their nonlinearity** causing them to evolve **without spreading out** in space, this behavior has characteristics **of particle interactions**. Soliton solutions arise naturally from the nonlinear Schrödinger equation.

Nonlinear Schrödinger equation

The **nonlinear Schrödinger equation (NLSE)** describes wave propagation in a **nonlinear, dispersive medium**. The nonlinear term arises from **mean field interactions** between adjacent atoms. The NLSE contains terms α, β, γ which are constants that change depending on the medium.

Wavefunction evolution $\rightarrow i\alpha \frac{\partial \psi}{\partial t} = -\beta \frac{\partial^2 \psi}{\partial x^2} + \gamma |\psi|^2 \psi$ $\leftarrow$ Nonlinear interaction term

Kinetic energy

The ' γ ' term tells us the type of **linearity** the system is in:

- If $\gamma < 0$ then it is self-focusing nonlinearity (bright solitons)
- if $\gamma > 0$ then its defocusing nonlinearity (dark solitons) (Fig 2.)

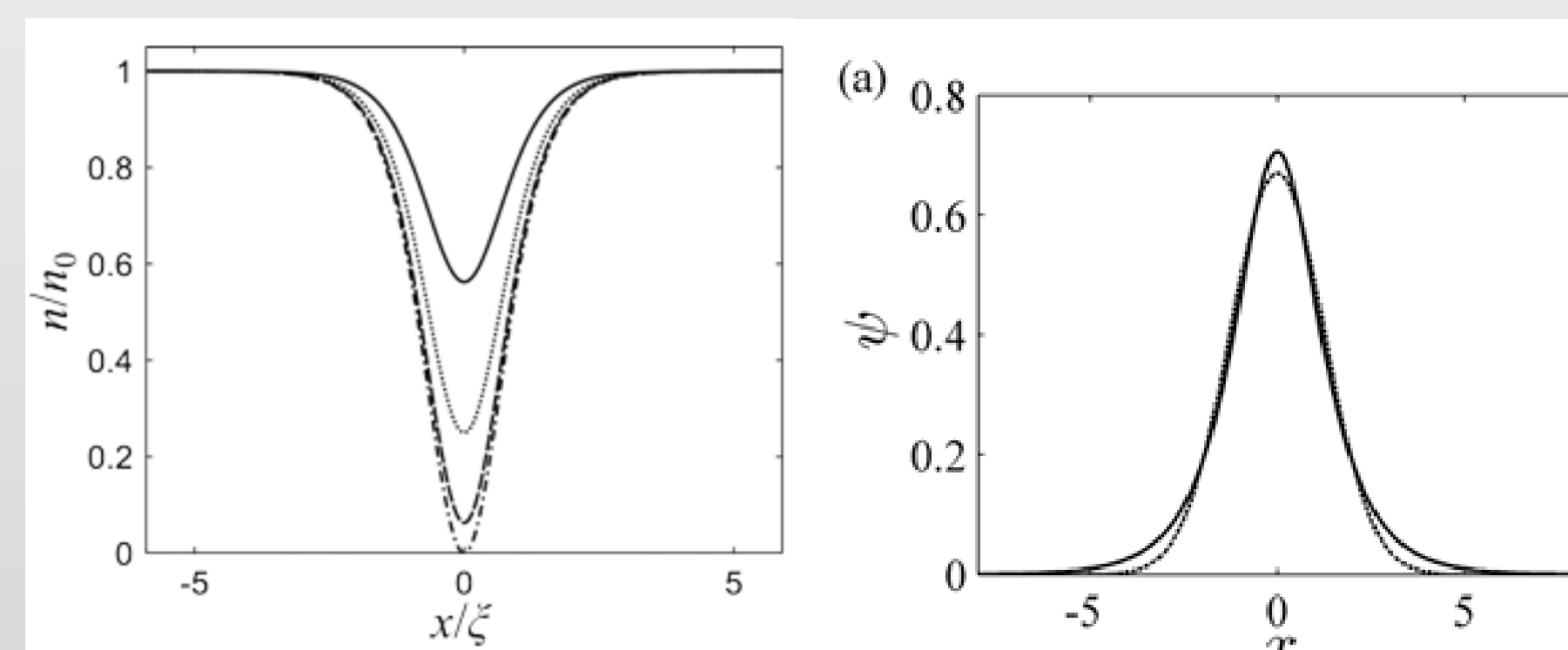


Fig 2. Density profile of dark (left) and bright (right) soliton solutions to the GPE [1].

Physical context and applications

When using the NLSE in terms of a BEC, the constant terms (α, β, γ) form the **GPE**:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + \frac{2\hbar^2 a_s}{ml_{\perp}^2} |\psi|^2 \psi$$

- Here, a_s is the **s-wave scattering length** and $l_{\perp}$ is the transverse harmonic oscillator length.
- We can model a BEC as a **single macroscopic wavefunction** ψ due to all bosons existing in the same quantum state at the same time.
- We use the **1D** form of the NLSE as the terms in 1D GPE still **preserve a lot of the important physics**
- **Physical applications include** [4] gravitational waves, optical fibres, nonlinear dynamics and integrable systems.

Semi-implicit finite difference method

To find soliton solutions to the GPE to obtain the density distribution, we use the method of semi-implicit finite difference (SIFD) :

- (1) Discretise the wavefunction with equidistant points on a N-point spacetime grid.
- (2) A Taylor series is then used to find approximation for second order solutions.

$$\left. \frac{\partial^2 \psi}{\partial x^2} \right|_{x=x_0} \approx \frac{\psi_{i+1}(t) + 2\psi_i(t) + \psi_{i-1}(t)}{(\Delta x)^2}$$

- (3) Rearrange the NLSE to get the time derivative at time t_0

$$\psi(t) \approx \frac{1}{(\Delta x)^2} A$$

Where 'A' is a tridiagonal matrix with 1,-2 and 1 terms.

- (4) Propagate the equation forward in time from a known initial state

$$\psi_i(t)(t_0 + \Delta t) \approx \psi_i(t_0) + \Delta t \left. \frac{\partial \psi_i}{\partial t} \right|_{t=t_0 + \Delta t/2}$$

- (5) Use a predictor corrector approach to approximate solutions which converge after, approximately 3 iterations

References

- [1] Barenghi, C.F. and Parker, N.G., 2016. *A primer on quantum fluids* (p. 121). Berlin: Springer.
- [2] Salasnich, L. (2017) 'Bright solitons in ultracold atoms', *Optical and Quantum Electronics*, 49(12). doi:10.1007/s11082-017-1247-5.
- [3] Ryōgo Hirota, Atsushi Nagai, J J C Nimmo and Gilson, C. (2004). *The direct method in soliton theory*. Cambridge, Uk ; New York: Cambridge University Press.
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